

# SHUBH RAJIV SUDAN

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## EDUCATION

### Rochester Institute of Technology

Master of Science in Artificial Intelligence; GPA: 3.9/4.0

Coursework:

Fall '25: Fundamentals of AI, Mathematical Methods of AI, Ethics of AI

Spring '26: NLP 1, Research Methods for AI, Fundamentals of ML, Entrepreneurship Mindset

Rochester, NY

Aug. 2025 – Present

### Symbiosis Institute of Technology

Bachelor of Technology in AI & ML; Honors in Cloud Computing

Pune, India

Aug. 2021 – Aug. 2025

## TECHNICAL SKILLS

Languages: Python, SQL, C++, JavaScript, Bash

Machine Learning: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, WGAN, CNN, LSTM, BERT, Transformers

AI Concepts: Generative Adversarial Networks (GANs), Deep Learning (CNN, LSTM), NLP, Computer Vision

Cloud & DevOps: AWS, Microsoft Azure, Google Cloud, Git/GitHub, Render

Web & Databases: Flask, REST APIs, React, Node.js, MySQL, MongoDB, SQLite

Analytics: Power BI, Microsoft Excel

## EXPERIENCE

### Research Assistant (College of Liberal Arts & Psychology)

Rochester Institute of Technology

Rochester, NY

Jan. 2026 – March 2026

- Conducting research on Argument Structure Construction, focusing on the intersection of computational linguistics and syntactic dependency.
- Developing a novel framework to calculate the Confidence Interval of Tagging, quantifying uncertainty in automated linguistic annotation to improve parser reliability.
- Analyzing large-scale corpora to refine tagging protocols, contributing to more robust confidence metrics for NLP models.

### Research and Development Intern

Diactio Technologies Pvt. Ltd.

Pune, India

Aug. 2024 – May 2025

- Led a team of 6 interns through the SDLC, collaborating with the CEO to align technical deliverables with business KPIs.
- Engineered an XGBoost classifier on Google Cloud to automate candidate screening, reducing manual review time by streamlining job role matching to their personality, deployed via Render.
- Integrated Google Vision OCR and OpenAI APIs into a medical mobile application to digitize patient records.
- Managed the procurement and configuration of 120+ mobile devices, negotiating vendor contracts to optimize logistics for the organization.

## PROJECTS

### Medical Image Augmentation using WGAN | Python, PyTorch, WGAN, CNN

Aug. 2025 – Dec. 2025

- Engineered a Wasserstein GAN (WGAN) to synthesize high-fidelity medical images, creating a "Gold Standard" augmentation pipeline for scarce datasets.
- Architected an optimized CNN classifier by doubling model depth and replacing LRN with Batch Normalization, achieving peak accuracy of 80%.
- Validated synthetic data efficacy by training a baseline model on WGAN-augmented samples, boosting baseline performance from 27% to 65%.

### Wind Turbine Damage Prediction | Python, TensorFlow, ResNet50, VGG-16

Jun. 2023 – Dec. 2023

- Developed a predictive maintenance system using Deep Learning to classify turbine blade defects (erosion, edge damage) from inspection imagery.
- Benchmarked performance of custom Sequential CNNs against Transfer Learning architectures (VGG-16, ResNet), achieving optimal results with ResNet.
- Optimized model evaluation using Precision, Recall, and F1-Score metrics to minimize false negatives in critical damage detection.

### Hinglish Chatbot Using BERT | Python, PyTorch, Transformers, NLP

Jun. 2023 – Dec. 2023

- Fine-tuned a BERT model on a large-scale Code-Mixed (Hinglish) corpus to engineer a context-aware conversational agent.
- Designed a custom Encoder-Decoder architecture to handle bilingual input sequences, achieving 80% accuracy in intent recognition tasks.
- Processed unstructured text data using Pandas and custom tokenizers to optimize training pipelines for localized language nuances.

## AWARDS & CERTIFICATIONS

### Best Presentation Award | AWARE AI Hackathon [GitHub Repository]

Feb. 2026

### Forward Learner Program | McKinsey.org (Business Strategy & Leadership)

Issued 2025